

TEST PLAN

OpenCart E-Commerce Application

Project Type: Manual QA Testing Portfolio

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This document defines the planned scope, approach, environment, execution process, and deliverables for manual testing of selected OpenCart e-commerce workflows.

Document Control

Field	Details
Document	OpenCart Test Plan
Purpose	Describe the approach for planning, designing, executing, and documenting manual tests for selected OpenCart functionality.
Status	Portfolio project document; update as the application and test coverage evolve.

1. Introduction

This test plan describes a practical manual testing exercise for an OpenCart e-commerce application. The project demonstrates QA documentation and test-design skills through test scenarios, detailed test cases, execution tracking, traceability, and a defect-reporting template. It is a portfolio exercise and does not imply access to production systems or formal business requirements.

2. Objectives

- Identify and document important user-facing e-commerce workflows.
- Design positive (Happy Path) and negative test cases.
- Validate behavior for navigation, account access, product discovery, cart, checkout, and order-related functions.
- Record actual results and execution status when tests are run.
- Document reproducible defects with evidence when an issue is observed.
- Maintain traceability from test scenarios to test cases.

3. Scope

3.1 In Scope

Module	Planned coverage
Homepage & Featured Products	Homepage availability, featured product display, product links and basic navigation.
Header & Navigation	Header controls, menus, account/cart navigation and links.
Currency	Currency selection and displayed price behavior across relevant pages.
User Registration	Mandatory fields, valid details, email validation, existing account checks, password confirmation and required terms.
Login & Authentication	Valid/invalid credentials, blank inputs, password masking and logout.
Wishlist & Product Compare	Adding/removing wishlist items, compare actions and relevant authentication behavior.
Shopping Cart	Adding/removing items, quantity changes, totals and empty-cart behavior.
Checkout	Checkout entry, billing/shipping information, shipping options, review and validation.
Product Search	Exact/partial searches, case variation, empty input, no-result searches and whitespace/special characters.
My Account	Account dashboard, customer details, address management and password-change validation.
Payment & Order Placement	Available payment flow, order submission, confirmation and duplicate-submission behavior where testable.

3.2 Out of Scope

- Source-code review and white-box testing.
- Formal security penetration testing.
- Load, stress, endurance, and other performance testing.
- Payment gateway certification or real financial transactions.

- Backend/database validation unless separately added.
- Full cross-browser/device coverage beyond browsers and viewport sizes actually checked.

4. Test Approach

- Functional testing: verify visible features behave according to their intended purpose.
- Positive testing: use valid data and expected user flows.
- Negative testing: check invalid, missing, or unsupported inputs and validation messages.
- UI and navigation checks: inspect labels, links, controls, and page transitions.
- End-to-end checks: follow representative journeys such as product discovery → cart → checkout.
- Regression considerations: recheck related workflows after a fix or relevant change.

5. Test Design Techniques

- Equivalence Partitioning for valid and invalid input groups.
- Boundary Value Analysis for quantity and field-length limits where defined by the application.
- Positive and negative test design.
- State-transition considerations for login/logout, cart updates, and order status where observable.
- Checklist-based UI and navigation verification.

6. Test Environment

Item	Planned setup / notes
Application	OpenCart e-commerce web application/demo instance used for the portfolio exercise.
Browser	Record the browser name and version actually used during execution.
Operating system	Record the operating system actually used during execution.
Tools	Web browser, Microsoft Excel, Git, and GitHub. Browser Developer Tools may be used where appropriate.
Test data	Use synthetic names, email addresses, addresses, and product selections. Do not commit passwords or personal information.

7. Test Data Management

- Use unique synthetic email addresses for registration and existing-user scenarios.
- Use clearly identified valid and invalid inputs for email, password, quantity, and address fields.
- Do not use real payment details or sensitive personal information.
- Keep reusable test data documented separately if the suite grows; never commit secrets to a public repository.

8. Entry Criteria

- The selected OpenCart instance is accessible.
- The relevant module or page is available for testing.
- Test scenarios and test cases have been reviewed for clarity.
- Required synthetic test data is prepared.
- Any account needed for a test is available, where permitted.

9. Exit Criteria

- Planned test cases have been executed where the environment allows.

- Each executed test has an actual result and status recorded.
- Observed defects are documented with reproduction steps and evidence where possible.
- Blocked or not-executed cases are identified with a reason.
- An execution summary records pass/fail/blocked/not-executed counts.

These are target criteria for this portfolio exercise, not a claim that all tests have already been executed.

10. Deliverables

- Test Plan (this document).
- QA workbook containing Summary, Master Test Cases, Test Scenarios, Test Execution, Traceability, and Bug Reports sheets.
- Execution results recorded as tests are performed.
- Bug reports and screenshots/evidence for issues actually observed.
- README describing project scope, structure, and limitations.

11. Defect Management

When an issue is observed, record a unique defect ID, concise summary, environment, preconditions, reproducible steps, expected result, actual result, severity/priority, status, and supporting evidence. Report only issues that were actually reproduced; do not invent defects to populate the portfolio.

12. Risks and Mitigations

Risk	Mitigation
Demo site availability or data resets	Record environment details and rerun affected cases when the site is available.
Features unavailable in the selected instance	Mark cases Blocked or Not Applicable with an explanation; do not assume behavior.
Dynamic prices, stock, or products	Record observed data and avoid hard-coding assumptions that may change.
Public repository exposure	Use synthetic data and remove credentials, personal information, and sensitive screenshots before publishing.

13. Assumptions and Limitations

- This is an independent portfolio project based on observable behavior of a selected OpenCart instance.
- Formal requirements, business rules, and internal implementation details may not be available.
- Traceability maps Test Scenarios to Test Cases; it is not a formal requirements-based RTM.
- The planned test-case count and coverage may change as scenarios are refined.
- Execution results must reflect actual test runs; planned tests must not be presented as executed.

14. Roles and Responsibilities

For this individual portfolio exercise, the author is responsible for understanding the selected workflows, preparing scenarios and test cases, executing tests where possible, recording results, documenting reproducible defects, and maintaining the repository.

15. Approval / Review

This document is prepared for demonstration and portfolio purposes. It can be reviewed and updated as test scope, test data, and application behavior evolve.